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A CONDITIONAL MULTISCALE HORIZONTAL CURVATURE CRITERION ON A FINITE GAUGE LATTICE

Main result: on $SU(N)^E$ with a gauge-invariant action, a lower bound $\kappa(t)$ on $Ric + 2\nabla^2 V_t$ in horizontal directions along the heat-kernel renormalization of the action gives $Var_\mu(f) \le 2\int_0^\infty e^{-2\int\kappa} dt \cdot \int |\nabla f|^2 d\mu$ for gauge-invariant f. For one SU(2) loop with density $1 + a\mathrm{Tr}(U)/2$ everything is explicit: $\kappa(t) = 1 - |a| e^{-3t/2}/(1 - |a|e^{-3t/2})$ and the constant is a hypergeometric function of $|a|$. The result is conditional and holds on one finite lattice at a time; no volume-uniform or continuum statement is made.

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Abstract

We state and prove a multiscale curvature criterion for Poincaré inequalities on a finite lattice gauge configuration space $M = SU(N)^E$. The action S is renormalized by the product heat semigroup, $V_t = -\log P_t(e^{-S})$, and the hypothesis is a lower bound $\kappa(t)$ on $\mathrm{Ric}_M + 2\nabla^2 V_t$ that is required only on vectors orthogonal to gauge orbits. The conclusion is a Poincaré inequality for gauge-invariant observables with constant $2\int_0^\infty \exp(-2\int_0^t \kappa)\,dt$. The bound $\kappa(t)$ may be negative at small scales. For the free action the criterion returns $2/N$. For one SU(2) loop with density $1 + a\cdot\mathrm{Tr}(U)/2$, $|a| < 1$, we compute everything in closed form: the optimal $\kappa(t)$, the time at which it becomes positive, and the constant $C(a) = (1 - |a|)^{-4/3}\,{}_2F_1(-4/3, 4/3; 7/3; |a|)$. In this example the criterion gives a finite constant for every $|a| < 1$, whereas the single-scale Bakry–Émery test stops at $|a| = 2/3$. We compare with the Holley–Stroock bound and with the numerically computed optimal constant, and we record that for the Wilson action on one loop the constant of the criterion grows by more than a factor of 30 per unit of β for $\beta \geq 1$, while the optimal constant decreases. The criterion is the adaptation to compact gauge manifolds of the multiscale Bakry–Émery method of Bauerschmidt and Bodineau; we have not found this adaptation in the literature, but we have not verified that it is new. The contribution is an explicit worked example and a precise conditional statement. No estimate uniform in the lattice, and no statement about the continuum limit, is proved.

Origins

This white paper develops the exercise “A conditional multiscale criterion on a finite gauge lattice” and its script `gauge_criterion.py`. The exercise tests one precise reading of the horizontal curvature criterion, Proposition 3 of the strong-coupling paper [8], at all scales of a renormalization flow instead of one.

Here	Exercise	Change
Definitions 1–8	Setting	Collected and numbered
Lemma 1: geometry and gauge invariance	Part 1	Proved
Lemma 2: flow identities	Part 2, first half	Two-parameter family $Q_{s,t}$ added for the proof
Lemma 3: gradient estimate	Part 2, second half	Proved; sharper form with $\|\nabla f\|$ in place of $\|\nabla f\|^2$
Lemma 4: dissipation identity	Part 3, identity	Proved
Theorem 5: the criterion	Part 3, boxed bound	Proved
Corollary 6: free action	Part 4	Comparison with the sharp constant added
Proposition 7: one SU(2) loop	Script and discussion	Closed form for $C(a)$ added
Proposition 8: comparison of bounds	Discussion	New here; corrects the role of the single-scale test
Remark 9: Wilson action	—	New here; numerical
Appendix A	Script	Reproduced with its output

0 Definitions

Definition 1 (configuration manifold). $\Gamma = (V, E)$ is a finite oriented graph, with $s(e)$ and $t(e)$ the source and target of $e \in E$. Fix $N \geq 2$. $M = SU(N)^E$ carries the product of the bi-invariant metrics $\langle X, Y\rangle = -\mathrm{Tr}(XY)$ on $\mathfrak{su}(N)$, and dU is the normalized Haar measure on M. $\Delta = \Sigma_e \Delta_e$ is the Laplace–Beltrami operator of M.

Definition 2 (gauge group and horizontal vectors). $\mathcal{G} = SU(N)^V$ acts on M by $(g\cdot U)_e = g_{s(e)} U_e g_{t(e)}^{-1}$. A function is gauge invariant if it is constant on $\mathcal{G}$-orbits. A vector $h \in T_U M$ is horizontal if it is orthogonal to the tangent space of the orbit $\mathcal{G}U$ at U.

Definition 3 (heat semigroup). $P_t = e^{t\Delta}$, $t \geq 0$. It acts on each edge by convolution with the heat kernel of $SU(N)$.

Definition 4 (action and measure). $S \in C^\infty(M)$ is gauge invariant, $Z = \int e^{-S} dU$, and $d\mu = Z^{-1} e^{-S} dU$.

Definition 5 (renormalized potential). $V_t = -\log P_t(e^{-S})$ and $d\mu_t = Z^{-1} e^{-V_t} dU$, so that $V_0 = S$ and $\mu_0 = \mu$.

Definition 6 (renormalization maps). For $0 \leq s \leq t$ and bounded f, $Q_{s,t} f = e^{V_t} P_{t-s}(f\, e^{-V_s})$, and $Q_t = Q_{0,t}$, that is $Q_t f = P_t(f\, e^{-S}) / P_t(e^{-S})$. The generator at scale t is $L_t = \Delta - 2\langle \nabla V_t, \nabla \cdot \rangle$.

Definition 7 (hypothesis (H) and the constant). $\kappa : [0, \infty) \to \mathbb{R}$ is locally integrable and, for every $t \geq 0$, every $U \in M$ and every horizontal $h \in T_U M$,

$$(\mathrm{Ric}_M + 2\nabla^2 V_t)(h, h) \geq \kappa(t)|h|^2. \qquad (H)$$

We write $\Lambda(t) = \int_0^t \kappa(s)ds$ and $C_\kappa = 2\int_0^\infty e^{-2\Lambda(t)} dt \in (0, \infty]$.

Definition 8 (one-loop model). Γ has one vertex and one edge, $N = 2$, and $\mathcal{G} = SU(2)$ acts by conjugation. We write $x = \tfrac{1}{2}\mathrm{Tr}\, U = \cos\theta$ with $\theta \in [0, \pi]$. For $|a| < 1$ the density is $w = 1 + ax$ with respect to Haar measure, that is $S = -\log(1 + ax)$.

1 Geometry and the flow

Lemma 1 (geometry and gauge invariance). (i) $\mathrm{Ric}_M = (N/2)\langle\cdot,\cdot\rangle$.

(ii) $\mathcal{G}$ acts on M by isometries. Consequently P_t commutes with the action of $\mathcal{G}$, V_t is gauge invariant for every t, and $Q_{s,t}f$ is gauge invariant whenever f is.

(iii) If $f \in C^1(M)$ is gauge invariant, then $\nabla f(U)$ is horizontal at every $U \in M$, including points with nontrivial stabilizer.

Proof. (i) For a bi-invariant metric on a compact Lie group, $\mathrm{Ric} = -\tfrac{1}{4}B$ with B the Killing form. On $\mathfrak{su}(N)$, $B(X,Y) = 2N\,\mathrm{Tr}(XY)$, so $\mathrm{Ric}(X,X) = -(N/2)\mathrm{Tr}\,X^2 = (N/2)|X|^2$. The Ricci tensor of a Riemannian product is the direct sum of those of the factors.

(ii) On each edge the action is a left translation by $g_{s(e)}$ composed with a right translation by $g_{t(e)}^{-1}$, both isometries of the bi-invariant metric. Isometries commute with Δ and therefore with P_t. Since S is invariant, $P_t(e^{-S})$ is invariant, hence so is V_t; the same argument applies to $P_{t-s}(f\,e^{-V_s})$.

(iii) If ξ is tangent to the orbit at U, then $\xi = (d/d\tau)(g_\tau \cdot U)$ at $\tau = 0$ for a curve g_τ in $\mathcal{G}$, and $\langle \nabla f, \xi\rangle = (d/d\tau) f(g_\tau \cdot U) = 0$. The argument uses only the orbit through U, so it does not require the orbit to have maximal dimension. ∎

Lemma 2 (flow identities). (i) V_t is smooth on $[0,\infty)\times M$ and $\partial_t V_t = \Delta V_t - |\nabla V_t|^2$.

(ii) Each $Q_{s,t}$ is linear, positivity preserving, $Q_{s,t}1 = 1$, $Q_{t,u}Q_{s,t} = Q_{s,u}$ for $s \le t \le u$, and $\int Q_{s,t}a\,d\mu_t = \int a\,d\mu_s$.

(iii) For smooth f, $\partial_t Q_{s,t}f = L_t Q_{s,t}f$ and $\partial_s Q_{s,t}f = -Q_{s,t}L_s f$.

Proof. Write $z_t = P_t(e^{-S}) = e^{-V_t}$. It is smooth and strictly positive because e^{-S} is, and $\partial_t z = \Delta z$. (i) From $\Delta e^{-V} = e^{-V}(|\nabla V|^2 - \Delta V)$ we get $\partial_t V = -\Delta z/z = \Delta V - |\nabla V|^2$.

(ii) Linearity and positivity are inherited from the heat semigroup. $P_{t-s}z_s = z_t$ gives $Q_{s,t}1 = 1$, and the semigroup property of P gives the composition rule. Since P_{t-s} preserves Haar integrals, $\int Q_{s,t}a\,d\mu_t = Z^{-1}\int P_{t-s}(a\,z_s)dU = Z^{-1}\int a\,z_s dU = \int a\,d\mu_s$.

(iii) Put $u = Q_{s,t}f$, so that $P_{t-s}(f\,z_s) = z_t u$. Differentiating in t, $\partial_t u = z^{-1}\Delta(zu) - (\Delta z/z)u = \Delta u + 2\langle \nabla \log z, \nabla u\rangle = L_t u$. For the second identity, the composition rule gives $Q_{s,t} = Q_{s+\varepsilon,t}Q_{s,s+\varepsilon}$ and $Q_{s,s+\varepsilon}f = f + \varepsilon L_s f + o(\varepsilon)$ by the first identity. ∎

2 Gradient estimate

Lemma 3 (gradient estimate). Assume (H). For every smooth gauge-invariant f and every $t \geq 0$,

$$|\nabla Q_t f| \leq e^{-\Lambda(t)} Q_t(|\nabla f|), \quad \text{hence} \quad |\nabla Q_t f|^2 \leq e^{-2\Lambda(t)} Q_t(|\nabla f|^2).$$

Proof. Fix t and $\varepsilon > 0$. For $r \in [0, t]$ let $G_r = Q_{0,r} f$, $w_r = (|\nabla G_r|^2 + \varepsilon)^{1/2}$ and $\psi(r) = Q_{r,t} w_r$. By Lemma 2(iii),

$$\psi'(r) = Q_{r,t}\big[-L_r w_r + \langle \nabla G_r, \nabla L_r G_r \rangle / w_r \big].$$

The Bochner identity for $L = \Delta - 2\langle \nabla V, \nabla \cdot \rangle$ reads

$$\tfrac{1}{2} L|\nabla G|^2 = |\nabla^2 G|^2 + \langle \nabla G, \nabla L G \rangle + (\mathrm{Ric} + 2\nabla^2 V)(\nabla G, \nabla G).$$

The chain rule for the diffusion operator L gives $Lw = L|\nabla G|^2/(2w) - |\nabla |\nabla G|^2|^2/(4w^3)$, and $|\nabla |\nabla G|^2|^2 = 4|\nabla^2 G(\nabla G)|^2 \leq 4|\nabla^2 G|^2 w^2$. Therefore

$$L_r w_r \geq [\langle \nabla G_r, \nabla L_r G_r \rangle + (\mathrm{Ric} + 2\nabla^2 V_r)(\nabla G_r, \nabla G_r)]/w_r.$$

By Lemma 1, G_r is gauge invariant and ∇G_r is horizontal everywhere, so (H) applies to the curvature term. This is the only place where the curvature tensor is evaluated, and it is evaluated on ∇G_r; this is why a horizontal bound suffices. We obtain

$$-L_r w_r + \langle \nabla G_r, \nabla L_r G_r \rangle / w_r \leq -\kappa(r)|\nabla G_r|^2/w_r \leq -\kappa(r) w_r + |\kappa(r)|\varepsilon^{1/2},$$

and, since $Q_{r,t}$ preserves positivity and constants, $\psi'(r) \leq -\kappa(r)\psi(r) + |\kappa(r)|\varepsilon^{1/2}$. Gronwall's inequality gives $\psi(t) \leq e^{-\Lambda(t)}\psi(0) + \varepsilon^{1/2} \int_0^t |\kappa(r)| e^{-(\Lambda(t)-\Lambda(r))} dr$. Here $\psi(t) = (|\nabla Q_t f|^2 + \varepsilon)^{1/2}$ and $\psi(0) = Q_t(|\nabla f|^2 + \varepsilon)^{1/2}$. Letting $\varepsilon \to 0$ gives the first inequality. The second follows from $(Q_t \varphi)^2 \leq Q_t(\varphi^2)$, valid because Q_t is a positive linear map with $Q_t 1 = 1$. ∎

The factor 2 in front of $\nabla^2 V_t$ comes from the drift of L_t. The argument is carried out on the smooth manifold M and never passes to the quotient $M/\mathcal{G}$, so configurations with nontrivial stabilizers need no separate treatment.

3 The criterion

Lemma 4 (dissipation identity). For smooth f and $t > 0$,

$$-(d/dt)\int_M (Q_t f)^2 \, d\mu_t = 2\int_M |\nabla Q_t f|^2 \, d\mu_t.$$

Proof. With $u = Q_t f$ and $z = z_t$, $Z\int u^2 d\mu_t = \int z\, u^2 dU$. Differentiating and using $\partial_t z = \Delta z$ and $\partial_t u = \Delta u + 2\langle \nabla \log z, \nabla u\rangle$,

$$(d/dt)\int z\, u^2 \, dU = \int [u^2 \Delta z + 2zu\Delta u + 4u\langle \nabla z, \nabla u\rangle] \, dU.$$

Integrating by parts on the compact manifold, $\int u^2 \Delta z = \int z \Delta(u^2) = \int z(2u\Delta u + 2|\nabla u|^2)$ and $\int 4u\langle \nabla z, \nabla u\rangle = 2\int \langle \nabla z, \nabla u^2\rangle = -2\int z\Delta(u^2) = -\int z(4u\Delta u + 4|\nabla u|^2)$. The terms in $u\Delta u$ cancel and the sum is $-2\int z|\nabla u|^2 dU$. ∎

Theorem 5 (conditional multiscale horizontal criterion). Assume (H). Then for every smooth gauge-invariant f,

$$\mathrm{Var}_\mu(f) \le C_\kappa \int_M |\nabla f|^2 \, d\mu, \qquad C_\kappa = 2\int_0^\infty \exp\!\left(-2\int_0^t \kappa(s)ds\right) dt.$$

The inequality is informative whenever $C_\kappa < \infty$; κ may be negative on a set of scales.

Proof. By Lemma 4, for $T > 0$,

$$\int f^2 d\mu - \int (Q_T f)^2 d\mu_T = 2\int_0^T \int |\nabla Q_t f|^2 d\mu_t \, dt.$$

M is compact and connected, so $P_T g \to \int g \, dU$ uniformly for continuous g. Hence $z_T \to Z$ and $Q_T f \to Z^{-1}\int f\, e^{-S} dU = \int f \, d\mu$ uniformly, and $\int (Q_T f)^2 d\mu_T \to (\int f \, d\mu)^2$. The left-hand side converges to $\mathrm{Var}_\mu(f)$. On the right, Lemma 3 and Lemma 2(ii) give

$$\int |\nabla Q_t f|^2 d\mu_t \le e^{-2\Lambda(t)} \int Q_t(|\nabla f|^2) d\mu_t = e^{-2\Lambda(t)} \int |\nabla f|^2 d\mu.$$

Integrating over $t \in [0, \infty)$ gives the claim. ∎

Corollary 6 (free action). If $S = 0$ then $V_t = 0$, (H) holds with $\kappa = N/2$ on all vectors, and $C_\kappa = 2/N$.

Proof. $P_t 1 = 1$, Lemma 1(i), and $2\int_0^\infty e^{-Nt} dt = 2/N$. ∎

The optimal Poincaré constant of Haar measure on M is $1/c_N = N/(N^2 - 1)$, the reciprocal of the smallest positive eigenvalue of $-\Delta$ [8, Lemma 1]. Since $N/(N^2 - 1) < 2/N$ for $N \ge 2$, the criterion is not sharp even in the free case: it reproduces the Bakry–Émery constant, not the spectral one. For $N = 2$ the two values are $2/3$ and 1.

What remains for an interacting action. For a given S one has $V_0 = S$, and (H) requires a lower bound on the horizontal Hessian of $V_t = -\log P_t(e^{-S})$ for every $t > 0$. If the conclusion is to be uniform over a family of lattices, one needs a common upper bound on $\int_0^\infty e^{-2\Lambda(t)} dt$ for the whole family. Theorem 5 does not provide such a bound.

4 One SU(2) loop in closed form

In the model of Definition 8 the gauge orbits are the conjugacy classes. At a point with $0 < \theta < \pi$ the orbit is a two-sphere and the horizontal space is the radial line; at $U = \pm\mathbb{1}$ the orbit is a point and every vector is horizontal. The metric restricted to the radial direction is $2d\theta^2$, Haar measure on class functions is $(2/\pi)\sin^2\theta\, d\theta$, $|\nabla x|^2 = (1 - x^2)/2$, and the characters $\chi_{n+1}(\theta) = \sin((n+1)\theta)/\sin\theta$ satisfy $\Delta\chi_{n+1} = -\tfrac{1}{2}n(n+2)\chi_{n+1}$.

Proposition 7 (one loop, exact). Let $b(t) = a\, e^{-3t/2}$. Then:

(i) $P_t w = 1 + b(t)x$ and $V_t = -\log(1 + b(t)x)$.

(ii) On the unit radial vector, $2\nabla^2 V_t$ equals $b(x+b)/(1+bx)^2$ with $b = b(t)$. Its infimum over M is $-|b|/(1-|b|)$, attained at the central element $x = -\mathrm{sgn}(b)$, where the Hessian is scalar.

(iii) The optimal function in (H) is $\kappa(t) = 1 - |b(t)|/(1 - |b(t)|)$. It is negative at $t = 0$ exactly when $|a| > \tfrac{1}{2}$, and then it becomes positive at $t^* = (2/3)\ln(2|a|)$.

(iv) $e^{-2\Lambda(t)} = e^{-2t}[(1 - |a|e^{-3t/2})/(1 - |a|)]^{4/3}$ and

$$C(a) = (1 - |a|)^{-4/3}\, {}_2F_1(-4/3, 4/3; 7/3; |a|) < \infty \text{ for every } |a| < 1.$$

(v) $C(0) = 1$, $C(0.6) = 1.964815\ldots$, and $(1 - |a|)^{4/3} C(a) \to \Gamma(7/3)^2/\Gamma(11/3) = 0.353328\ldots$ as $|a| \to 1$.

Proof. (i) $x = \tfrac{1}{2}\chi_2$ and $\Delta\chi_2 = -(3/2)\chi_2$. (ii) With $V = -\log(1 + b\cos\theta)$, $V_\theta = b\sin\theta/(1 + b\cos\theta)$ and $V_{\theta\theta} = [b\cos\theta(1 + b\cos\theta) + b^2\sin^2\theta]/(1 + b\cos\theta)^2 = b(x+b)/(1+bx)^2$. The unit radial vector is $2^{-1/2}\partial_\theta$, so $2\nabla^2 V$ on it equals $V_{\theta\theta}$. For $b > 0$, $(d/dx)[b(x+b)/(1+bx)^2] = b(1 - bx - 2b^2)/(1+bx)^3$. If $b < \tfrac{1}{2}$ this is positive on $[-1, 1]$ and the minimum is at $x = -1$. If $b \geq \tfrac{1}{2}$ the function increases up to $x = (1 - 2b^2)/b$ and decreases afterwards, so its minimum over $[-1, 1]$ is the smaller of the endpoint values $-b/(1-b)$ and $b/(1+b)$, again the value at $x = -1$. The case $b < 0$ follows from $x \mapsto -x$. At $U = \pm\mathbb{1}$ the conjugation action fixes U and acts irreducibly on the tangent space, so the Hessian of an invariant function is a multiple of the metric, equal to its radial limit.

(iii) $\mathrm{Ric} = 1$ for $N = 2$. At regular points only the radial direction is horizontal, and at central points the Hessian is scalar, so the infimum in (H) is $1 - |b|/(1-|b|)$. It is negative when $|b| > \tfrac{1}{2}$. (iv) With $db = -(3/2)b\, dt$, $\int_0^t |b|/(1-|b|)ds = (2/3)\ln[(1-|b(t)|)/(1-|a|)]$, which gives $e^{-2\Lambda}$. The substitution $u = e^{-3t/2}$ turns $C = 2\int_0^\infty e^{-2\Lambda}dt$ into $(4/3)(1-|a|)^{-4/3}\int_0^1 u^{1/3}(1 - |a|u)^{4/3}du$, and Euler's integral gives $\int_0^1 u^{1/3}(1-|a|u)^{4/3}du = (3/4)\,{}_2F_1(-4/3, 4/3; 7/3; |a|)$. (v) Evaluation, and Gauss's formula for ${}_2F_1$ at 1. ∎

For $a = 0.6$: $\kappa(0) = -0.5$, $t^* = (2/3)\ln 1.2 = 0.1215\ldots$, and $C = 1.9648$. A lower bound that is negative at the initial scale is compensated by the positive curvature recovered after the heat flow has smoothed the density.

Proposition 8 (comparison of bounds on one loop). For gauge-invariant f and the density $1 + ax$:

(i) Single-scale Bakry–Émery. $\mathrm{Ric} + \nabla^2 S \geq 1 - |a|/(2(1 - |a|))$ on horizontal vectors, with equality at $x = -\mathrm{sgn}(a)$. The test gives the constant $[1 - |a|/(2(1 - |a|))]^{-1}$ for $|a| < 2/3$ and gives nothing for $|a| \geq 2/3$.

(ii) Holley–Stroock. The constant is at most $(2/3)(1 + |a|)/(1 - |a|)$ for every $|a| < 1$.

(iii) Multiscale. $C(a)$ of Proposition 7 is finite for every $|a| < 1$ and grows like $(1 - |a|)^{-4/3}$.

Proof. (i) The generator of the Langevin dynamics of μ is $\Delta - \langle \nabla S, \nabla \cdot \rangle$, whose Bakry–Émery tensor is $\mathrm{Ric} + \nabla^2 S$; by Proposition 7(ii) with $t = 0$ its horizontal infimum is $1 - |a|/(2(1 - |a|))$, and the horizontal form of the argument is Proposition 3 of [8]. (ii) On class functions the Poincaré constant of Haar measure is $2/3$, and the density lies between $1 - |a|$ and $1 + |a|$; the perturbation lemma of Holley and Stroock [9] multiplies the constant by at most the ratio. (iii) Proposition 7. ∎

$\lvert a\rvert$	Multiscale $C(a)$	Single-scale	Holley–Stroock	Optimal (numerical)
0	1.0000	1.0000	0.6667	0.6667
0.3	1.2548	1.2727	1.2381	0.6526
0.5	1.6217	2.0000	2.0000	0.6269
0.6	1.9648	4.0000	2.6667	0.6088
2/3	2.3259	fails	3.3333	0.5947
0.8	3.9214	fails	6.0000	0.5615
0.9	8.6962	fails	12.667	0.5323
0.95	20.509	fails	26.000	0.5165
0.99	166.21	fails	132.67	0.5033

The last column is the reciprocal of the smallest positive eigenvalue of the Dirichlet form on class functions, computed in a basis of 80 characters; it is a numerical value, not a proved bound.

Three observations. First, at $a = 0.6$ the single-scale test does not fail: it gives the constant 4. What is negative at $a = 0.6$ is $\kappa(0)$, the initial value of the multiscale bound, which involves $2\nabla^2 S$ instead of $\nabla^2 S$. The single-scale test fails from $|a| = 2/3$ onwards, and there the multiscale criterion still gives a finite constant. Second, the multiscale constant is smaller than the single-scale one at every tabulated $|a| > 0$ where both exist. Third, no bound is close to the optimal constant, which decreases as $|a|$ grows while all three bounds increase; close to $|a| = 1$ the Holley–Stroock bound, of order $(1 - |a|)^{-1}$, is better than the multiscale one, of order $(1 - |a|)^{-4/3}$.

Remark 9 (Wilson action on one loop; numerical). For $S = -\beta \operatorname{Tr} U = -2\beta x$ the flow is no longer closed in one character. Here $\kappa(0) = 1 - 2\beta$ exactly. Computing V_t from 400 characters and $\kappa(t)$ on a grid of angles gives:

β	$\kappa(0)$	$\min_t \kappa(t)$	Multiscale constant	Single-scale	Optimal (numerical)
0.1	0.80	0.80	1.13	1.11	0.665
0.25	0.50	0.50	1.38	1.33	0.656
0.5	0.00	0.00	2.08	2.00	0.625
1	−1.00	−1.05	6.55	fails	0.527
2	−3.00	−4.48	222	fails	0.328
3	−5.00	−10.2	1.7×10^4	fails	0.208
4	−7.00	−18.4	1.9×10^6	fails	0.146

For the Wilson action the criterion remains finite at every β but its constant grows very rapidly, by factors of 34, 78 and 108 per unit of β between $\beta = 1$ and $\beta = 4$, while the optimal constant decreases; for $\beta \le \tfrac{1}{2}$ it is slightly worse than the single-scale test. The loss comes from the neighbourhood of $U = -\mathbb{1}$, where the action is strongly non-convex and the measure is exponentially small. A flow that smooths every edge at the same rate cannot separate that region from the rest. These are numerical values for one loop and are not proved bounds.

Remark 10 (research question). Can the restriction to horizontal directions give a lower bound in (H) that is uniform in the volume for an interacting plaquette model in a range of couplings where the bound on the full Hessian fails? The one-loop calculation shows how to pose the comparison. It does not answer it.

5 Status

Result	Status	Scope and remaining input
Lemmas 1–4	Proved	Any finite graph, any $N \geq 2$
Theorem 5: criterion	Proved, conditional on (H)	One finite lattice at a time; (H) must be verified for the action
Corollary 6: free action	Proved	Not sharp: $2/N$ against $N/(N^2 - 1)$
Proposition 7: one SU(2) loop	Proved	Density $1 + a\cdot\mathrm{Tr}(U)/2$ only
Proposition 8: comparison	Proved	Optimal constants in the table are numerical
Remark 9: Wilson action	Numerical	One loop; no proof
(H) for an interacting lattice action, uniformly in the volume	Open	Remark 10
Novelty of Theorem 5	Not verified	Adaptation of [1, 2]; not found in [3–6], which were not all read in full
Continuum limit and mass gap	Not addressed	—

Context. Bauerschmidt and Bodineau introduced the multiscale Bakry–Émery criterion along the Polchinski flow for measures on linear spaces [1, 2]. Shen, Zhu and Zhu proved log-Sobolev and Poincaré inequalities for lattice Yang–Mills at strong coupling by the single-scale method [3]; Cao, Nissim and Sheffield obtained an area law for $\beta < 1/(8(d-1))$ by a dynamical argument [4]; Piazza and Shen proved a log-Sobolev inequality for two-dimensional SU(N) lattice Yang–Mills at every coupling by a two-scale argument [5]. The present paper does not improve any of these results.

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Appendix A. The script gauge_criterion.py

The script implements the exercise for the one-loop model. It applies the heat semigroup exactly in the character basis, uses the analytic $\kappa(t)$ of Proposition 7, and checks numerically, for the observable $f = x + x^2/4$, the gradient estimate of Lemma 3 at 45 scales, the dissipation identity of Lemma 4, and the inequality of Theorem 5. It requires NumPy.

```
"""Finite gauge lattice exercise: one vertex, one SU(2) loop, conjugation gauge action.

x = Tr(U)/2 = cos(theta). The radial metric is 2 dtheta^2 and Haar
measure is (2/pi) sin(theta)^2 dtheta. Use a positive density w=1+a*x
(action S=-log(w)), and polynomial class observables. Requires NumPy.
"""
import argparse

import numpy as np
from numpy.polynomial import Polynomial as P
from numpy.polynomial.legendre import leggauss

X = P([0, 1])
G = (1 - X**2) / 2  # |grad x|^2


def heat(poly, t):
    """Apply exp(t Delta) in the SU(2) character basis, without spatial gridding."""
    chars = [P([1]), 2 * X]
    while len(chars) <= poly.degree():
        chars.append(2 * X * chars[-1] - chars[-2])
    remainder = poly
    result = P([0])
    for n in range(poly.degree(), -1, -1):
        char = chars[n]
        coeff = remainder.coef[n] / char.coef[n] if n <= remainder.degree() else 0
        result += coeff * np.exp(-t * n * (n + 2) / 2) * char
        remainder = remainder - coeff * char
    return result


def run(a, f):
    if not abs(a) < 1:
        raise ValueError("The density requires |a| < 1.")
    theta = np.arange(1, 513) * np.pi / 513
    x = np.cos(theta)
    weights = 2 * np.sin(theta)**2 / 513
    w = 1 + a * X
    grad2 = G * f.deriv()**2
    mean = weights @ (w(x) * f(x))
    variance = weights @ (w(x) * (f(x) - mean)**2)
    energy = weights @ (w(x) * grad2(x))

    # For b=a exp(-3t/2), twice the horizontal Hessian of V=-log(1+b*x)
    # is b*x/(1+b*x) + b^2*(1-x^2)/(1+b*x)^2. Its minimum occurs
    # at the adverse endpoint. At either central element isotropy makes
    # the Hessian scalar. Adding Ric=1 gives kappa=1-|b|/(1-|b|).
    def factor(s):
        # exp(-2 integral_0^t kappa), with s=exp(-t)
        return s**2 * ((1 - abs(a)*s**1.5)/(1 - abs(a)))**(4/3)

    nodes, quadrature = leggauss(96)
    s = (nodes + 1)/2
    constant = quadrature @ (factor(s)/s)  # 2 dt = -2 ds/s

    def moment(t):
        d = heat(w, t)(x)
        n = heat(w*f, t)(x)
        return weights @ (n*n/d)

    worst_excess = 0.
    worst_residual = 0.
    for t in np.geomspace(0.001, 8, 45):
        d, n = heat(w, t), heat(w*f, t)
        u_prime = (n.deriv()(x)*d(x) - n(x)*d.deriv()(x))/d(x)**2
        lhs = G(x)*u_prime**2
        rhs = factor(np.exp(-t))*heat(w*grad2, t)(x)/d(x)
        worst_excess = max(worst_excess, np.max(lhs-rhs))
```

```
        h = 1e-5
        derivative = (moment(t+h)-moment(t-h))/(2*h)
        worst_residual = max(worst_residual, abs(derivative + 2*(weights @ (d(x)*lhs))))
    print(f"a={a:g} C={constant:.9g} Var={variance:.9g} C*energy={constant*energy:.9g}")
    print(f" gradient excess={worst_excess:.3g} dissipation residual={worst_residual:.3g}")
    if max(variance-constant*energy, worst_excess, worst_residual) > 1e-6:
        raise ArithmeticError("Numerical criterion check exceeded tolerance.")
    if a == 0 and abs(constant-1) > 1e-12:
        raise ArithmeticError("Free normalization failed.")


if __name__ == "__main__":
    parser = argparse.ArgumentParser(description=__doc__)
    parser.add_argument("--coupling", type=float, default=0.6)
    args = parser.parse_args()
    observable = X + 0.25 * X**2
    run(0, observable)
    if args.coupling:
        run(args.coupling, observable)
```

Output of `python gauge_criterion.py --coupling 0.6`:

```
a=0 C=1 Var=0.25390625 C*energy=0.390625
 gradient excess=0 dissipation residual=1.49e-10
a=0.6 C=1.96481549 Var=0.25015625 C*energy=0.841186632
 gradient excess=0 dissipation residual=2.85e-10
```

Coupling a	Criterion constant C	Measured variance	Poincaré upper bound
0	1.000000	0.253906	0.390625
0.6	1.964815	0.250156	0.841187

The free case recovers $C = 2/N = 1$. In both runs the sampled gradient estimate has no positive violation and the largest dissipation residual is 2.85×10^{-10}. These checks cover the one-loop model and one observable; they illustrate the statements proved above and are not a substitute for the proofs.

Repository, citation and licence

GitHub repository: https://github.com/teknoquants/A-CONDITIONAL-MULTISCALE-HORIZONTAL-CURVATURE-CRITERION-ON-A-FINITE-GAUGE-LATTICE
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```

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